



Senan Jayasinghe

AI & ML Engineer

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Data Science undergraduate with experience working on AI/ML projects and building intelligent solutions. Enjoys working with data, experimenting with machine learning models, and exploring how AI can be applied to real world problems. Interested in developing practical AI applications and continuously improving through projects. A quick learner who adapts well, works well with others, and enjoys taking on new challenges.

TECHNICAL SKILLS

Languages: Python, R, Java, TypeScript, C#, C++

Full Stack : Spring Boot, .NET, MERN, Redux TK, FastAPI, RESTAPI, WebSockets

AI & ML: TensorFlow, PyTorch, Scikit-learn, Hugging Face, LangGraph, RAG Pipelines

Data & NLP: NumPy, Pandas, Matplotlib, NLTK, TF-IDF, OpenCV

Database: SQL, MongoDB, PostgreSQL, Oracle DB, Qdrant

Cloud & DevOps: Docker, Ci/CD, GCP, AWS

Tools: GitHub, Git, Jupyter Notebook, Power BI, Tableau, VS Code, Visual Studio, Postman, SSMS, n8n, Excel

PROFESSIONAL SKILLS

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| • Agile teamwork | • Critical Thinking |
| • Problem Solving | • Strong Communication skills |
| • Time management under deadline | • Adaptability and flexibility |
| • Analytical Thinking | • Research & Knowledge Sharing |

EDUCATION

- **BSc (Hons) in Information Technology Specializing in Data Science (UG)** | Nov 2023 – 2027
Sri Lanka Institute of Information Technology (SLIIT) | CGPA: 3.30 / 4.00
- **G.C.E. Advance Level 2022 (Science Stream)**
Sivali Central College Ratnapura | Passed With 2 S 's and a C

PROFESSIONAL EXPERIENCE

Freelance Backend Developer – Upwork (Third Eye Trading)

- Developed and optimized C# backend services, implementing real-time communication features using SignalR and WebSockets for efficient live data synchronization.
- Improved live data update performance by 30% and collaborated with the React.js frontend team to deliver integrated full-stack functionality.

PROJECTS

Arfy AI – Personal Voice Assistant (Python, PyTorch, LangChain, LangGraph, Qdrant, RAG, faster-whisper, PyQt6, OpenGL, FastAPI) - [GitHub](#)

Built an AI-powered voice assistant that understands user commands and performs tasks such as web search, weather updates, app control, and media actions. Integrated Generative AI (LLMs, RAG) and AI agent workflows to make responses more contextual and useful.

CampusMind AI – University Learning Assistant (Python, FastAPI, Qdrant, BM25, Gemini, Groq, Cohere, RAG, Next.js) – [GitHub](#)

Developed an AI-powered university learning assistant using a hybrid RAG pipeline with semantic and keyword search, intelligent multi-LLM routing, enabling context-aware question answering, document analysis, and automated study material generation.

PawCare – AI Powered Cat Monitoring System (Python, PyTorch, YOLOv8, ResNet18, FastAPI, WebSockets, OpenCV, Electronjs, TypeScript) – [GitHub](#)

Developed a real-time computer vision monitoring platform integrating object detection, identity recognition, and breed classification. Built a high-concurrency FastAPI backend with asynchronous PostgreSQL storage and WebSocket-based telemetry streaming for live camera analytics and automated alerts.

Stock Prediction Model (Python, Tensorflow, NumPy, Pandas, yfinance) - [GitHub](#)

Developed an LSTM-based stock forecasting system using historical financial data. Performed data pre processing, feature engineering, and time-series analysis to identify market patterns and predict price movements through a Streamlit dashboard.

Bangkok Airbnb Market Intelligence (Python, Pandas, DuckDB, XGBoost, Streamlit, Hugging Face) - [GitHub](#)

Developed an end-to-end data engineering and ML pipeline for analyzing 28K+ Airbnb listings. Built a reproducible ETL pipeline, star schema warehouse, statistical hypothesis testing, XGBoost price prediction, customer segmentation, multilingual sentiment analysis and an interactive dashboard for market insights.

CareerPulse AI – Job Tracking & Automation System (Docker, n8n, APIs, LLMs, Supabase, GmailAPI)

Developed an AI-powered automation system that collects job listings, filters relevant opportunities, scores them using an LLM, and sends real-time alerts through Telegram and email. Used Dockerized n8n workflows, APIs, and Supabase for automation and data storage

CERTIFICATIONS

- AI & ML SLIIT
- MongoDB Data Modeling Path
- Introduction to LangGraph – Python

ADDITIONAL ACTIVITIES

- National Level Chess Player
- Singer & Instrumentalist

REFERENCES

- **Tiranga De Alwis** - Associate Software Engineer | Riskconnect, Inc
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- **Samadhi Chathuranga Rathnayaka** - Lecturer (Department of Computer Science) | SLIIT
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